Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/01/2022
PI Name:	Li, Chen Ph.D.		
Project Title:	Impacts of Gravity on Dropwise Condensation-Enhanced Heat Pipes		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	29208	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Physical Sciences NNH20ZHA001C
Start Date:	06/01/2020	**End Date:**	05/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2022 (original end date was 5/31/2021) per NSSC information (Ed., 1/20/22)		
Key Personnel Changes/Previous PI:	Dr. Cassandra J. Runyon serves as the South Carolina NASA EPSCoR Director; she is the administrative Principal Investigator on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Chao, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0153		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project Primarily consisting of evaporation and condensation sections, the heat pipe is a light, highly efficient, and durable two-phase device that transports heat over a distance with a small temperature drop and a type of primary devices in thermal control systems of satellites, space shuttles, and crewed space stations. Ultrahigh-performance, long, and light-weight heat pipes are critical to future space missions (e.g., crewed missions to Mars) and hence, highly desired. Numerous efforts have been made to improve heat pipe performance by engineering the evaporation section and are nearly approaching the physical limits. However, the heat pipe condensation section that results in an order of magnitude higher thermal resistance than the evaporation one has not been well developed. It is promising if a performance leap of heat pipe technologies can be realized by enhancing the condensation section. Heat transfer rate of dropwise condensation (DWC) is 10 times higher than film wise condensation adopted in existing heat pipe technologies. Most recently, using super-durable Ni-graphene coatings in a terrestrial environment, the Science Principal Investigator's team has successfully demonstrated 8 times higher effective thermal conductivity by promoting DWC inside heat pipes. However, the enhancement was highly sensitive to orientations. In most space environments, due to the reduced gravity, droplets would attach on the wall and DWC could totally fail without quickly removing droplets. The objective of this proposed project is to explore a feasibility in implementing DWC for space missions. Conducting experiments in a drop tower would be the first step to verify the feasibility of wick designs in rapidly removing droplets inside heat pipes. In this proposal, condensation sections of heat pipes will be coated with the demonstrated Ni-graphene coatings. Moreover, various wicking structures will be designed to eventually realize rapid droplet removal in a microgravity environment. The gap between the Ni-graphene coated wall and wicking structures would play a critical role in maximizing DWC and hence, heat pipe performances. Heat pipes with various wicking structure designs and gaps will be evaluated before and in drop tower experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The heat pipe or vapor chamber, which consists of an evaporator and a condenser, has been widely used in thermal management of electronics. However, their performance has been fundamentally limited by the physical nature of filmwise condensation (FWC). In this project, we have successfully demonstrated the effectiveness of novel Ni-Gr coatings in promoting dropwise condensation (DWC) and drastically enhancing heat pipe performance nearly an order of magnitude compared with the one with conventional heat pipes with FWC. The new heat pipe technologies can be utilized to reduce the temperature difference between the heating components such as CPUs or GPUs and coolants (air or liquid coolants), which would effectively address the bottle neck imposed by thermal management in developing next generation electronics that demand high performance. Equally important, the technology developed in this project can be extended to develop highly conductive vapor chambers that are widely used as heat spreaders for high-power devices.
Task Progress:	In the year of 2021, we have successfully demonstrated that grooved copper heat pipes with Nickle-graphene (Ni-Gr) coatings achieved extremely high effective thermal conductivity (keff). Grooves are the simplest wicking structures and used in heat pipes for space missions. However, this type of heat pipe cannot be evaluated against gravity to complete our proposed research. Due to the travel restriction that resulted from COVID-19, the drop tower test in the NASA Glenn Research Center (GRC) could not be carried out. Scientifically, it would be a sound alternative method to demonstrate the operation of the new heat pipe technology in against gravity in the Principal Investigator's (PI's) research lab. In the year of 2022, we focused on developing various wicking structures that can facilitate dropwise condensation promoted by the Ni-Gr coatings, and more importantly, enable operations of heat pipes in the vertical direction to again 1g gravity. The challenge lies in the gap design, which should be large to leave room for nucleation of droplets and at the same time, be small to be able to absorb droplets. The first design was to fabricate this type of wicking structures on the inner surface of circle copper tubes, but failed due to the difficulty to manage the gap with a relatively uniform thickness. Dropwise condensation was not promoted in several samples, which led to very poor performance. To address the associated fabrication challenge, we decided to fabricate the same type of wicking structures on flat heat pipes. It is more likely to realize a uniform gap thickness using micro pillars. The flat heat pipes have successfully operated with dropwise condensation promoted by the Ni-Gr coatings as we expected. Its thermal resistance is only 1/40 of pure copper. More importantly, the flat heat pipe shows insensitive to orientations. Three positions were experimentally evaluated: horizontally up, down, and vertically with the evaporator on the top. The thermal resistance on the vertical direction is slightly higher than those obtained in other two directions. Without conducting drop tower tests, we still successfully demonstrated that the Ni-Gr promoted dropwise condensation can work in a microgravity environment.
Bibliography Type:	Description: (Last Updated: 12/21/2023)
Articles in Peer-reviewed Journals	Huang G, Li W, Zhong G, Abdulshaheed AA, Li C. "Optimizing L-shaped heat pipes with partially-hybrid mesh-groove wicking structures." Int. J. of Heat and Mass Transfer. 2021 May;170:120926. https://doi.org/10.1016/j.ijheatmasstransfer.2021.120926 , May-2021